

Digital Material Passport

Manufacturer		Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com		Bridge Constructors Europe Ltd. Engineering Plaza 456 75001 Paris FR procurement@bridgeconstructors.example.com	Order	PO-98765 · 20000 kg Pos. 5 · 2025-04-20
			Delivery	DN-12345 · 20000 kg Pos. 1 · 2025-05-19
Certificate Receiver Civil Infrastructure Authority Government Building 789 Administration Wing 1000 Brussels BE inspections@infra-authority.example.gov.be				

Product

Product Name	Structural Steel S420N Plate	Name (EN)	1.8902
Batch ID	H-45678-01	Shape	Plate - Width 2500 mm - Thickness 25 mm - Length 12000 mm
Surface Condition	Shot blasted and primed	Product Norm	EN 10025-3 (2019)
Weight	20000 kg		EN 1090-2 (2018)
Production Date	2025-05-18		
Country of Origin	DE		

Chemical Analysis

Heat Number H-45678 - Melting Process EAF+LF+VD - Casting Method ContinuousCasting - Casting Date 2025-05-17 - Sample Location Ladle									
Symbol	C	Mn	Si	P	S	Nb	V	Ti	CEV
Unit	%	%	%	%	%	%	%	%	%
Min		1.00							
Max	0.22	1.70	0.60	0.025	0.020	0.050	0.050	0.030	0.48
Actual	0.16	1.38	0.32	0.016	0.008	0.022	0.034	0.009	0.40
CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.4%									

Mechanical Properties

Tensile Strength ¹	Symbol	1	2	3	Average	Min	Max	Spec Min	Spec Max	Method	Status
Value [MPa]	Rm	548	550	552	550	548	552	520	680	EN ISO - 6892-1	✓
Yield Strength ¹	Symbol	1	2	3	Average	Min	Max	Spec Min	Method	Status	
Value [MPa]	ReH	438	440	442	440	438	442	420	EN ISO - 6892-1	✓	
Elongation after - fracture	Symbol	1	2	3	Average	Min	Max	Spec Min	Method	Status	
3 specimens tested, gauge length 5.65#S ₀ , transverse to rolling direction											
Value [%]	A	20.5	21	21.5	21	20.5	21.5	19	EN ISO - 6892-1	✓	
Charpy V-notch - Impact Energy	Symbol	1	2	3	Average	Min	Max	Std Dev	Spec Min	Method	Status
3 specimens tested at -20°C, transverse to rolling direction											
Value [J]	KV	63	65	67	65	63	67	2	40	EN ISO - 148-1	✓

¹ 3 specimens tested at 20°C, transverse to rolling direction

Supplementary Tests

Test	Result / limits	Method	Status
Ultrasonic Testing Class S2E2	Yes - No recordable indications exceeding acceptance criteria	EN 10160	✓
Through-thickness Properties	Z25	EN 10164	✓

Weldability

Yes - Satisfactory welding properties

Internal Method based on EN ISO -
15614-1



Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10025-3:2019 and EN 10204:2004 type 3.1. The product complies with the Construction Products Regulation (EU) No 305/2011 and is suitable for use in structural applications according to EN 1090-2:2018, up to and including Execution Class EXC3.



Validated by **John Smith**, Quality Manager, Quality Assurance - 2025-05-20

Validated by **Maria Schmidt**, Quality Manager, Quality Assurance - 2025-05-20